problem, but we can see all the work the LLM is doing to reason out the problem.

Open up all three prompts: `decompose_steps.jinja2`, `calculate_steps.jinja2`, and `calculate_solution.jinja2` (see listings 10.13, 10.14, and 10.15, respectively). All three prompts shown in the listings can be compared to show how outputs chain together.

Listing 10.13 `decompose_steps.jinja2`

```
system:
You are a problem solving AI assistant.
Your job is to break the users problem down into smaller steps and list
the steps in the order you would solve them.
Think step by step, not in generalities.
Do not attempt to solve the problem, just list the steps. #1

user:
{{statement}}     #2
```

#1 Forces the LLM to list only the steps and nothing else
#2 The initial problem statement

Listing 10.14 `calculate_steps.jinja2`

```
system:
You are a problem solving AI assistant.
You will be given a list of steps that solve a problem.
Your job is to calculate the output for each of the steps in order.
Do not attempt to solve the whole problem,
just list output for each of the steps.     #1
Think step by step.     #2

user:
{{statement}}

{{steps}}     #3
```

#1 Requests that the LLM not solve the whole problem, just the steps
#2 Uses the magic statement to extract reasoning
#3 Injects the steps produced by the decompose_steps step